

Revision & Anti-Fabrication Audit Report

Companion document to “A Trustworthy, Explainable, and Auditable AI Assistant for DeFi Personal-Finance Risk Assessment” (IEEE-format paper, v2)

Prepared for supervisor review · Generated 27 August 2026 · Paper build: 19 pages, US Letter

1. Purpose and Scope

This report documents the supervisor-driven revision of the IEEE paper and provides a complete evidence trail for every quantitative claim it contains. Its purpose is to let a reviewer confirm two things independently: first, that each result, figure, table, and statistic in the paper is traceable to a concrete source artifact from the project, and second, that the revision strengthened rigor and disclosure without inventing data, inflating results, or removing unfavorable findings. The report also records one figure that was deliberately excluded, with the exact numerical reason, and lists the fields that still require the author's input.

Nothing in the paper was generated to make the system look stronger than the artifacts support. Where the evidence is unfavorable to the deployed model — the guardrail dominates most decisions, a simpler baseline matches or edges the neural network, the labels are proxies rather than outcomes — the paper states so, and this report shows where each such statement comes from.

2. Summary of the Revision

Four explicit supervisor requirements were addressed, alongside a broader tightening of the methodology, statistics, references, and research-question mapping. Their status is summarized below.

#	Supervisor requirement	How it was met in v2	Status
1	Separate per-model confusion matrices from actual predictions, with error analysis.	Fig. 4 shows all four classifiers' confusion matrices on the identical 600-case holdout; §IV-C reads the error structure (adjacent-band only; zero Low ↔ High confusion; High-band recall compared across models).	Done
2	Real statistical validation, preserving any non-significant result.	Bootstrap 95% CIs, 5-fold CV, McNemar tests with Holm–Bonferroni, per-class + macro/weighted metrics, and effect sizes (Tables VI–VIII). The non-significant MLP-vs-LR result is retained and foregrounded.	Done
3	RQ → objective → evidence mapping with honest status labels.	Table XI maps RQ1–RQ6 to method and evidence with Supported / Partially supported labels (no automatic “achieved”). Three are only partially supported, by design.	Done
4	Architecture diagram matching the actual code.	Fig. 1 separates the offline training pipeline from online inference, shows the leakage-aware feature contract (Eq. 5), and routes features exactly as the notebook does (5 raw inputs to the classifier; 3 ratios to clustering).	Done
—	Broader rigor: leakage disclosure, guardrail-dominance quantification, ROI correction, reference verification.	§III-D quantifies residual arithmetic leakage; §IV-F quantifies guardrail dominance; §III-K / Table IX disclose and correct the ROI indexing error; all 22 references were checked for consistency and IEEE formatting.	Done

All five rows reflect changes present in the delivered v2 .docx / .pdf.

3. Metric Provenance

Every headline number in the paper maps to a named artifact file and field, or to a value computed directly from one. The core artifacts are: `finance_risk_metrics_final.json` and `model_comparison.csv` (model metrics, CV, McNemar, bootstrap CIs), `finance_risk_metrics.json` (the MLP holdout confusion matrix), `cv_validation.csv` (cross-validation), and the project notebook `finalnb.ipynb` (clustering indices, guardrail sweep, and the deterministic simulation). Rounding to the precision shown in the paper is the only transformation applied.

Paper location	Value(s) as printed	Source artifact → field / derivation
Abstract; Table II	3,000 records; 944 unique users (~95.8% repeat)	<code>classification_dataset_with_risk_labels.csv</code> — row count vs. unique <code>user_id</code> count.
§IV-B; Table V	Silhouette 0.370	<code>finance_risk_metrics_final.json</code> → <code>cluster_silhouette_full</code> = 0.37020.
§IV-B	Calinski–Harabasz 2158.37; Davies–Bouldin 0.925; ARI 0.868 ± 0.064	<code>finalnb.ipynb</code> clustering-evaluation cell (CH, DB) and resampling-stability cell (ARI mean $\pm$ std).
Abstract; §IV-C; Table VII	Accuracy 0.910; macro-F1 0.881	<code>...metrics_final.json</code> → <code>holdout_test_accuracy</code> 0.91, <code>holdout_test_f1_macro</code> 0.880765.
Table VI (per-class)	Low .922/.971/.946; Med .935/.878/.906; High .761/.823/.791	Computed from MLP confusion matrix in <code>finance_risk_metrics.json</code> <code>holdout_confusion_matrix</code> [[235,7,0],[20,260,16],[0,11,51]]; accuracy = 546/600.
Fig. 4 (4 matrices)	MLP/LR/RF/DT confusion matrices	Per-model holdout predictions from the saved bundle; MLP panel equals the JSON matrix. Cells reproduce the accuracy/macro-F1 of Table VII exactly.
Table VII	MLP .910/.881/.9859/.9514; LR .892/.864/.9861/.9520; RF; DT	<code>model_comparison.csv</code> (<code>test_accuracy</code> , <code>test_macro_f1</code> , <code>roc_auc_ovr_macro</code> , <code>pr_auc_macro</code>).
Table VIII (McNemar)	vs LR $\chi^2_{2.70}$ (24/13) $p=.100$; vs RF $\chi^2_{1.83}$ (22/13) $p=.176$; vs DT $\chi^2_{48.67}$ (82/13) $p<.001$	<code>...metrics_final.json</code> → <code>mcnemar_checks</code> (<code>chi2</code> , discordant counts, <code>approx_p_value</code>).
§IV-C, §IV-E	Bootstrap 95% CI: acc [0.888, 0.933]; macro-F1 [0.847, 0.912]	<code>...metrics_final.json</code> → <code>selected_model_accuracy_ci_95</code> , <code>selected_model_macro_f1_ci_95</code> .
§IV-C, §IV-E	5-fold CV: acc 0.912 ± 0.009 ; macro-F1 0.867 ± 0.018	<code>cv_validation.csv</code> — MLP <code>cv_accuracy</code> 0.912333 ± 0.009226 , <code>cv_macro_f1</code> 0.866533 ± 0.017966 .
Table VIII text	Odds ratios 1.85, 1.69, 6.31	Derived from the discordant-pair counts (b/c): 24/13, 22/13, 82/13.
§IV-F; Fig. 7	Guardrail: Low 5.87% / Med 33.53% / High 60.60%; $\geq$ cluster 100%; strictly > 84.13%; model-can-change 0.00 / 5.87 / 39.40%	<code>finalnb.ipynb</code> guardrail-analysis cell: Eq. 13 rule + reconciliation applied across the labeled dataset.
Table IX; Fig. 8	Corrected ROI 25.72 / 15.40 / 2.43%; original 20.18 / 12.34 / 2.22%; drawdowns -3.79 / -2.08 / 0.00%	<code>finalnb.ipynb</code> deterministic 12-month simulation; corrected value divides by initial capital (Eq. 18), original by month-1 (the off-by-one).

File names abbreviated with “...” share the `finance_risk_metrics_final.json` prefix. No value in the paper lacks a row here.

4. Figure Inventory and One Deliberate Exclusion

Eight figures are included. Figs. 1 and 4 were rebuilt for this revision (architecture and all-model confusion matrices); the remainder are the project's own vetted outputs, reused unchanged or regenerated from verified counts. Each figure's numbers were checked against the artifacts in Section 3.

Fig.	Content	Provenance / check
1	End-to-end architecture (training vs. inference)	Rebuilt to match finalnb.ipynb feature routing and the Eq. 5 contract.
2	Correlation heatmap of engineered quantities	Project figure; surplus $\leftrightarrow$ expense -0.89 , income $\leftrightarrow$ surplus 0.81 (shared algebra).
3	K-means cluster separation (PCA + ratio space)	Project figure; silhouette 0.370 matches Table V.
4	Confusion matrices, all four models	Rebuilt from holdout predictions; reproduces Table VII accuracy/macro-F1.
5	Per-class ROC & PR curves (MLP)	Per-class OVR AUCs average to ≈ 0.986 , consistent with Table VII macro-OVR.
6	Comparative bars (Acc/F1/ROC-AUC/PR-AUC)	Regenerated from model_comparison.csv values.
7	Guardrail dominance (share + model influence)	From the §IV-F guardrail sweep.
8	Deterministic 12-month ROI, original vs. corrected	From Table IX values.

4.1 The excluded all-models ROC/PR figure — and why

An additional figure overlaying ROC and precision–recall curves for all four models was available but was deliberately left out, because the AUC values it would display are computed under a different averaging convention than Table VII and reverse the paper's honest finding. This is worth recording in full, because it is exactly the kind of silent inconsistency the revision was meant to eliminate.

The project stored two different area-under-curve summaries for each model:

- Macro one-vs-rest (used in Table VII): model_comparison.csv $\rightarrow$ roc_auc_ovr_macro. MLP = 0.9859, LR = 0.9861.
- Micro-averaged (used by the all-models curve figure): ...metrics_final.json $\rightarrow$ curve_auc.micro_roc_auc. MLP = 0.9888, LR = 0.9813.

These two conventions order the two leading models in opposite directions:

Averaging	MLP ROC-AUC	LR ROC-AUC	Ordering	MLP PR-AUC	LR PR-AUC
Macro-OVR (Table VII)	0.9859	0.9861	LR > MLP	0.9514	0.9520
Micro (excluded fig.)	0.9888	0.9813	MLP > LR	0.9786	0.9630

Macro-OVR weights every class equally; micro pools all decisions and is dominated by the well-represented Low/Medium bands.

For a three-class problem with a 9.2% minority (High risk), the macro one-vs-rest average is the appropriate and more conservative summary: it does not let the easy majority classes mask minority-class behavior. It also happens to be the less flattering choice for the deployed MLP, since under it logistic regression marginally edges the MLP on both AUCs — the finding the paper reports in §IV-D and Table VII. Publishing the micro-based all-models curve alongside Table VII would have shown the MLP ahead on AUC, directly contradicting the text. The figure was therefore excluded and the macro-OVR convention used consistently throughout. Per-class curves for the deployed MLP are still shown (Fig. 5), whose per-class AUCs average to the same macro value.

5. Statistical Validation and the Preserved Non-Significant Result

The comparison the supervisor specifically asked to be reported honestly — MLP versus logistic regression — is not statistically significant, and the paper says so rather than engineering an apparent win. On the 600-case holdout, McNemar's test gives $\chi^2 = 2.70$ on 24/13 discordant pairs ($p = 0.100$); against the random forest $\chi^2 = 1.83$ ($p = 0.176$); only the decision tree is beaten significantly ($\chi^2 = 48.67$, $p < 0.001$). A Holm–Bonferroni correction across the three comparisons leaves every decision unchanged. The bootstrap intervals overlap heavily and five-fold CV agrees with the holdout. Accordingly the paper frames the close comparisons as “no demonstrated difference,” explicitly an absence of demonstrated superiority rather than proof of equivalence, and retains the MLP for its probability outputs (which the occlusion explainer consumes) rather than on any claim of superior accuracy.

6. Preserved Honest Findings and Limitations

None of the following was softened or removed to flatter the system; each is stated in the paper with its source in Section 3 above.

- **Guardrail dominance (high severity).** The deterministic guardrail escalates 60.6% of users to High risk and is $\geq$ the cluster signal for 100% of users, so the learned model decides the outcome for only a minority. The 91% accuracy is therefore a property of one component, not of end-to-end decision quality.
- **Residual arithmetic leakage (high).** Two of the three clustering features (surplus, expense_ratio) are exact algebraic functions of the classifier inputs, so column-level separation does not remove leakage; only debt_pressure is truly withheld. This is disclosed, not concealed.
- **Proxy labels, no ground truth (high).** Labels come from clustering, not recorded default/loss outcomes; no real-world predictive validity is claimed.
- **Non-significant model selection (medium).** The MLP is not statistically better than logistic regression, which edges it on AUC.
- **Backtest indexing error (medium).** The original ROIs were inflated by an off-by-one; corrected values are used and both are shown side by side (Table IX).
- **Entity leakage in the split (medium).** A row-level split over 944 unique users among 3,000 rows can place a user in both partitions, so generalization may be overstated.
- **Façade layer / simulated ledger (low–med).** Part of the source tree is a not-yet-extracted presentation layer, and the SHA-256 audit ledger is in-memory, not a deployed chain.

7. Anti-Fabrication Statement

To the standard set for this revision: no experiments, results, statistical values, confusion matrices, references, datasets, or methodological details were invented. Every quantity in the paper reproduces a value in the artifacts listed in Section 3, transformed only by rounding to the displayed precision. No numerical value was altered to favor the system, and no unfavorable result was removed — the guardrail-dominance, leakage, non-significance, and ROI-correction findings are reported prominently. The 22-entry reference list is the author-supplied verified list and was not substituted with machine-generated citations; the review only checked internal consistency, completeness of in-text citations, and IEEE formatting. No claim is made that the paper is guaranteed to be accepted.

8. Outstanding Items (Require Author Input)

Four identity fields on the title page remain intentionally unfilled and are the only blockers to a final submission copy:

- Author name(s)
- Roll number(s)
- Session
- Supervisor name

Provide these and they will be inserted into the front matter; no other content change is pending.